MI1-T-85075A(AS)
31 December 1985
SUPERSEDING
MIL-T-85075(AS)
14 November 1977

MILITARY SPECIFICATION

TIE DOWN ASSEMBLY, HELICOPTER BLADE, GENERAL SPECIFICATION FOR

This specification is approved for use within the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the tie down assemblies for use on helicopter blades.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extend specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-365	Silver Plating, Electrodeposited, General Requirements for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Air Engineering Center, Systems Engineering and Standardization Department (Code 93), Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document. or by letter.

AMSC N/A FSC 1730

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SPECIFICATIONS (Continued)

MILITARY

MIL-C-5541	Chemical Conversion Coatings on Aluminum Alloys
MIL-H-6875	Heat Treatment of Steel, Process for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-10727	Tin Plating, Electrodeposited or Hot-dipped, for Ferrous and Nonferrous Metals
MIL-C-26074	Coatings, Electroless Nickel, Requirements for
MIL-T-85075/1	Tie Down Assembly, He licopter Blade, CH-46 Helicopter
MIL-T-85075/2	Tie Down Assembly, He licopter Blade, CH-53 Helicopter

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-794	Part and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals

MILITARY HANDBOOKS

MIL-HDBK-5	Aerospace Vehicle Structures, Metallic Materials and Elements for

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(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

9633-78	Electrodeposited Coatings of Zinc on Iron and Steel, Specification for
AMS 5643	Steel Bars, Forgings, Tubing and Rings, Corrosion Resistant

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified in the contract or purchase order (see 4.3 and 6.2.1), the contractor shall furnish sample tie down assemblies for first article inspection(s) and approval. A new contractor, supplier, or manufacturer shall be required to complete a first article inspection (4.3).

3.3 Materials. Materials used in the manufacture of tie down assemblies shall be of aircraft quality, suitable for the purpose, and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used, provided they are equivalent to Government specifications with respect to operating characteristics and that a savings in weight or cost can be accomplished. Contractor's specifications must be satisfactory to the Government and contain provisions for adequate tests as specified herein. The use of contractor's specifications will not constitute waiver of Government inspection.

3.3.1 Metals. All metals shall be compatible with the intended temperature, functional, service and storage conditions to which the components will be exposed. The metals shall possess adequate corrosion

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resistant characteristics or shall be suitably protected by the use of coatings equivalent to those listed in Table I to resist corrosion which may result from such conditions as dissimilar metal combinations, moisture, and salt spray as applicable. Where not indicated, class or type is at the option of the manufacturer, subject to approval by the procuring activity. Dissimilar metals are defined in MIL-STD-889. Unless otherwise specified herein, physical properties of all metals shall meet the minimum requirements of MIL-HDBK-5.

3.3.1.1 Copper and aluminum alloys. All copper alloys except for bearing surfaces shall be suitably protected with a coating selected from Table I, except that cadmium and zinc plating shall not be used in applications subject to abrasion. Other metallic coatings, which shall have been demonstrated to be satisfactory to the Government, such as electrodeposited 85 percent tin and 15 percent cadmium alloy, may be used. All aluminum alloys shall be anodized in accordance with MIL-A-8625, type II coating, except that in the absence of abrasive conditions the anodize may be type I coating, where applicable, or may be a chemical film in accordance with MIL-C-5541.

TABLE I. Metal Coatings.

Coating	Specification
Cadmium plating	QQ-P-416, type II, class 2
Zinc plating	ASTM-B633-78, type II, SC3
Chromium plating	QQ-C-320
Nickel plating	QQ-N-290
Silver plating	QQ-S-365
Tin plating	MIL-T-10727, type I
Electroless nickel	MIL-C-26074

3.3.2 Plastic parts. The use of plastic parts shall be subject to the approval of the procuring activity for the specific application involved.

3.3.3 Selection of materials, parts and processes. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD- 43, except as provided in the following paragraph.

3.3.3.1 Standard parts Standard part (MS, AN or Specification sheet) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their (MS, AN or Specification sheet) part numbers. Commercial utility parts, such as screws, bolts, nuts, and cotter pins may be used, provided they possess suitable properties and are replace-able by the standard parts (MS, AN or Specification Sheet) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor’s drawings. In the event there is no suitable corresponding standard part in effect on date of invitation of bids, commercial parts may be used provided they conform to all requirements of this specification and the detail specification.

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3.4 Design and construction. The design and construction of the tie down assembly shall be as specified herein and in accordance with the applicable specification sheet.

3.4.1 Dimensions and tolerances. Dimensions and tolerances shall be as specified on the applicable specificati on sheet.

3.5 Performance. Tie down assembl ies shall be performed satisfactorily when subjected to the applicable tests specified in this specification, and the applicable specification sheets.

3.6 Finish. All surfaces shall have a smooth finish and be free from burrs and sharp edges.

3.7 Identification of product.

3.7.1 Marking. Tie down assembly marking shall be as indicated on the applicable specification sheet, MIL-T-85075/1 or MIL-T-85075/2.

3.7.2 Part numbering and drawing numbering. The manufacturer’s basic part number and drawing number shall be the same.

3.8 Workmanship. Workmanship shall be high grade to insure that tie down assemblies are of uniform quality and free from burrs, slivers, sharp edges, or other defects which would affect their service or performance, and shall be uniform in their appearance without any mold flash.

4. QUALITY ASSURANCE PROVISIONS

4.1 Repsonsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection and testing of tie down assemblies shall be classified as:

a. First article inspection (4.3)

b. Quality conformance inspection (4.4)

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4.3 First article inspection. It shall be mandatory to require all new manufacturers or first-time contractors to successfully complete first article inspection.

4.3.1 Samples. Samples for first article testing shall consist of six tie down assemblies.

4.3.2 Inspections. First article inspections shall consist of the tests specified under 4.6 conducted on tie down assembly samples, in the order specified in Table II.

Table II. First article inspections.

Test	Para.	Defect
Examination of products	4.6.1	Critical
Shackle pull test	4.6.2	Critical
Load test	4.6.3	Critical
Drop test	4.6.4	Critical
Engagement-disengagement test	4.6.5	Critical
Salt fog test	4.6.6	Critical

4.3.3 First article test report. The contractor shall prepare a report of the first article tests in accordance with MIL-STD-831 showing that the tie down assemblies tested conform to this specification. Two copies shall be furnished to the procuring activity and two additional copies shall be forwarded with the verification test samples. First article test report shall be in accordance with 6.3.1.

4.3.4 Verification test samples for the procuring activity. The contractor shall submit to the agency specified in 6.2(d), together with the first test report, 5 sample tie down assemblies to be used as follows:

a. For review of the mechanical construction of the product.

b. To perform any tests included in this specification, after reviewing the contractor’s test report.

4.4. Quality conformance inspection. The inspection requirements specified herein are classified as follows:

a. Individual inspection (4.4.2)

b. Sampling inspection (4.4.3)

The quality conformance test report shall be in accordance with 6.3.1.

4.4.1 Inspection lot. An inspection lot shall consist of tie down assemblies of a Particular size, and material produced under essentially the same manufacturing conditions and presented for inspection at the same time.

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4.4.2 Individual inspection.

4.4.2.1 Visual examination. Each tie down assembly shall be visually examined to meet the requirements of 4.6.1. Any tie down assembly failing to pass the examination shall be rejected.

4.4.2.2 Operational examination. Each tie down assembly shall be subjected to 5 engagement-disengagement cycles of the engagement-disengagement test (4.6.5). Any tie down assembly failing to meet the requirements of 4.6.5 shall be rejected.

4.4.3 Sampling inspection. A random sample shall be selected from each inspection lot in accordance with MIL-STD-105, Inspection Level S-1, and shall be subjected to the load test (4.6.3) in the vertical position only. If this sample fails the load test defined in 4.6.3, the inspection lot represented by the sample shall be rejected.

4.5 Test conditions.

4.5.1 Environment. Unless otherwise specified, all testing shall be done at room temperature (60°F - 90°F).

4.5.2 Test fixture. Test fixtures to be used in 4.6.2 and 4.6.3, shall be in accordance with the requirements of the applicable specification sheet.

4.6 Test methods.

4.6.1 Examination of product. Tie down assembly shall be thoroughly examined to determine conformance to the physical and dimensional requirements of this specification and the applicable specification sheet.

4.6.2 Shackle pull test. One tie down assembly shall be subjected to a horizontal force of 100 lbs. (See Figure 1) at a point just below the body of the tie down assembly. No separation of the shackle from the body shall occur.

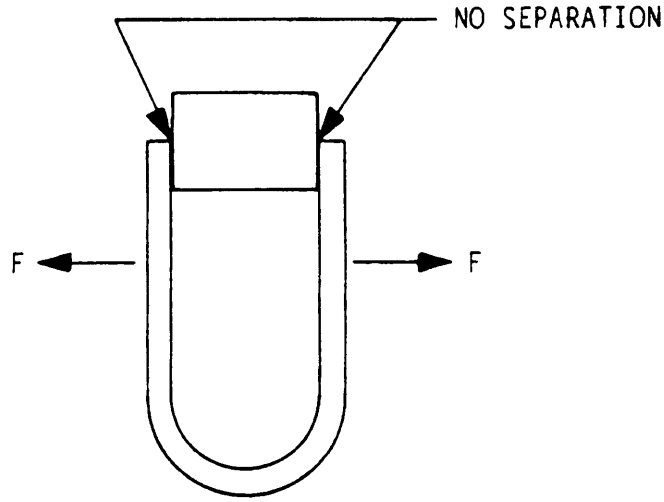


Figure 1.

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4.6.3 Load test. One tie down assembly shall be engaged in a rigidly mounted test fixture (see applicable specification sheet). Test loads shall be applied to the shackle in 200 lb. Increments to the total load specified on the applicable specification sheet, at each of the three positions shown in Figure 2. The point of contact of the locking device in the test fixture shall remain constant throughout the test. The tie down assembly and test fixture shall exhibit no evidence of cracking, shearing, distortion, or other damage resulting from the test.

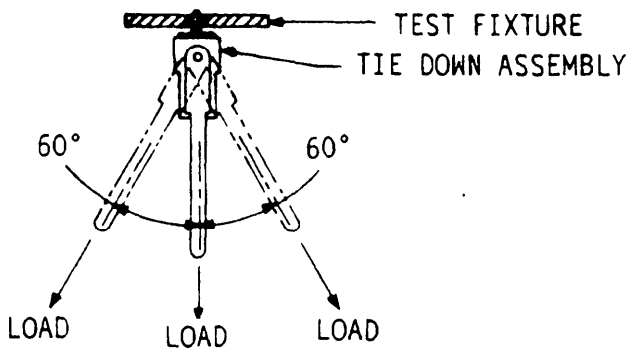


Figure 2.

4.6.4 Drop test. One tie down assembly shall be dropped from a height of twenty feet onto a flat concrete surface a minimum of twenty times. The same tie down assembly shall be dropped onto 1/2-inch steel plate a minimum of twenty times. There shall be no evidence of cracking, chipping, distortion or other damage to the tie down assembly which would affect its service. The drop test specimen shall be subjected to the engagement-disengagement test (4.6.5) as specified.

4.6.5 Engagement-disengagement test. The test specimen shall be engaged, then disengaged from a rigidly mounted test fixture (see applicable specification sheet) one thousand times, unless otherwise specified. The mechanical operation of the test specimen shall be smooth and positive, and no deformation of the tie down assembly or test fixture shall result. There shall be no deformation of the release ring.

4.6.6 Salt fog test. One tie down assembly shall be subjected to a salt fog test procedure in accordance with Method 509.1 of MIL-STD-810. The tie down assembly shall be checked before removing any salt deposits and again after rinsing with water. There shall be no evidence of pitting, cracking, or conversion of components, and the specimen shall be subjected to the engagement-disengagement test (4.6.5) for one hundred cycles.

4.7 Noncompliance. If a sample fails to pass a critical inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or

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both, as warranted, and on all units of product which can be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action has been taken. After the corrective action has been taken, the inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.8 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-STD-1299, MIL-STD-130, and MIL-STD-794.

5. PACKAGING

5.1 Preservation and packaging. Tie down assemblies shall b preserved and packaged in accordance with commercial methods unless level A is specified (See 6.2).

5.1.1 Level A. Tie down assemblies shall be preserved and packaged in accordance with MIL-STD-794.

5.1.2 Commercial. Preservation and packaging shall be sufficient to afford physical protection against damage during shipment and handling from the supply source to the Government using activity for immediate use.

5.2 Packing. Tie down assemblies shall be packed in accordance with commercial methods unless level A or B is specified (See 6.2).

5.2.1 Level A or B. When level A or B is specified, packaged items shall be packed for overseas or domestic shipment, respectively, to comply with MIL-STD-704.

5.2.2 Commercial. Tie down assemblies packaged as specified in 5.1 shall be packed In a manner to insure carrier acceptance and safe delivery destinations. Containers shall conform to the rules and regulations applicable to the mode of transportation.

5.3 Marking. Marking of interior and exterior shipping containers shall be in accordance with MIL-STD-129 and MIL-STD-130.

6. NOTES

6.1 Intended use. Tie down assemblies covered by this specification are intended to be used in securing helicopter blades to landing surfaces. Specific helicopters shall use tie down assemblies specified on applicable specification sheets.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Specification sheet part numbers.

c. Applicable levels of preservation, packaging, and packing (See section 5).

d. Where the first article verification test samples should be sent and instructions concerning the submittal of the test reports (see 4.3.4). First article inspection (see 4.3) shall be required of all new contractors or manufacturers.

e. Engineering change procedure in accordance with NAVAIR INST 4275.3A. The Naval Air Systems Command, Department of the Navy, Washington, DC 20360 (Attn: AIR-534) will be responsible for review of Class 1 changes under the contract. The Commanding Officer, Naval Air Engineering Center, Lakehurst, NJ 08733-5100 (Attn: SEED) will be responsible for approval of Class II changes.

6.2.1 First article inspection. Acquisition documents should specify the fol lowing provisions for the first article inspection.

6.2.1.1 State if first article inspection is required, and indicate:

a. Where the first article inspection is to be conducted (at the con tractors plant or Government or commercial laboratory).

b. That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fullfill all other requirements of the specification and contract.

c. Sample tie down assemblies subjected to first article inspection shall not be considered, accepted or shipped as part of the contract or order.

6.2.2 Quality conformance inspection. After a contractor has successfully completed a first article inspection or is in continuous production of the tie down assembly, the quality conformance inspection shall be required. The data obtained from the first article inspection(s) shall be used as reference for the follow-on quality conformance inspection(s).

6.3 Data. For the information of Contractors and Contracting Officers, any of the data specified in (a) applicable documents listed in Section 2 of this specification, or (b) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or purchase order. The data to be furnished shall be listed in DD Form 1423 (Contractor Data Requirement List), which shall be attached to and a part of the contract or order.

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6.3.1 Contract data requirement. Items of deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirements	Applicable DID
4.3.3	First Article Test Report	DI-T-5329
4.4	Quality Conformance Test Report	DI-T-5329

6.4 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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